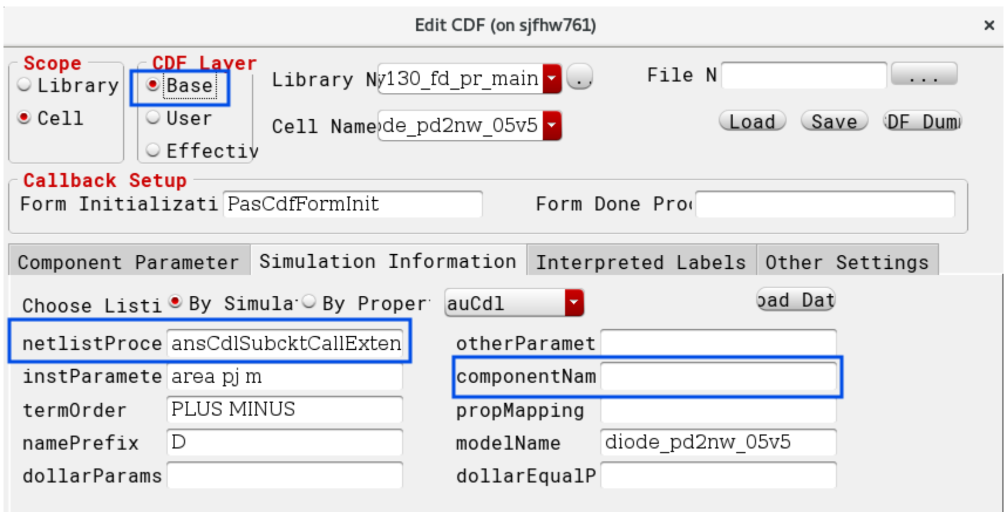
Sky130 LVS: Required Changes July 23, 2025

The following changes are required to achieve correct LVS results.
The changes must be applied to the primary Sky130 PDK library: sky130_fd_pr_main
When making CDF changes, be sure to choose "Base" under "CDF Layer" in the Edit CDF form.

Cell Type: all diodes

Change required: CDF
Under Simulation Information, auCDL must be updated:

netlistProcedure:	ansCdlSubcktCallExtended
componentName:	should be empty



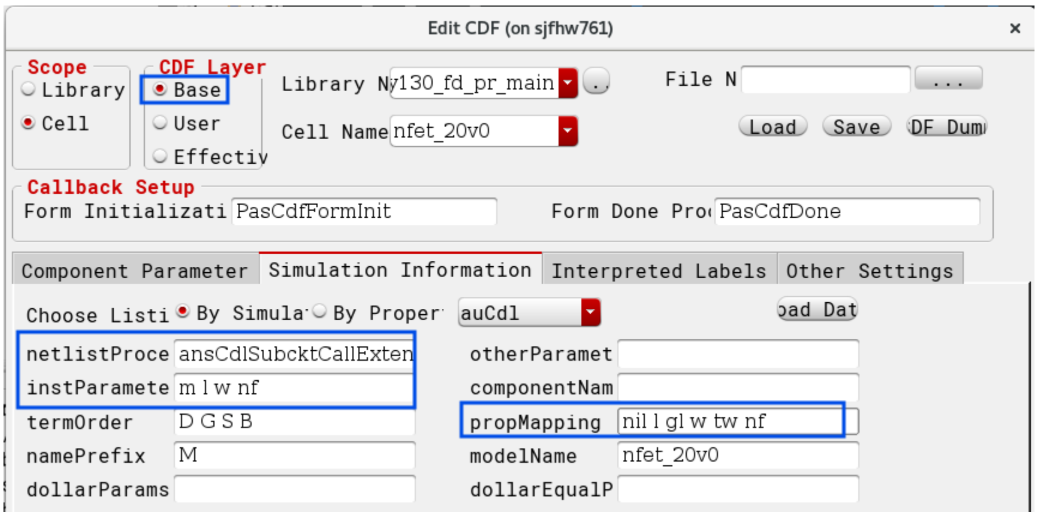
Cell Type: HV MOS

All 3 HV MOS devices must be updated: nfet_20v0, nfet_20v0_zvt, pfet_20v0

Change required: CDF

Under Simulation Information, auCDL must be updated:

netlistProcedure:	ansCdlSubcktCallExtended
componentName:	should be empty
instParameters:	“nf” added
propMapping:	“nf” added



Cell Type: BJT devices

All 3 BJT devices must be updated: npn_05v5 / npn_11v0 / pnp_05v5

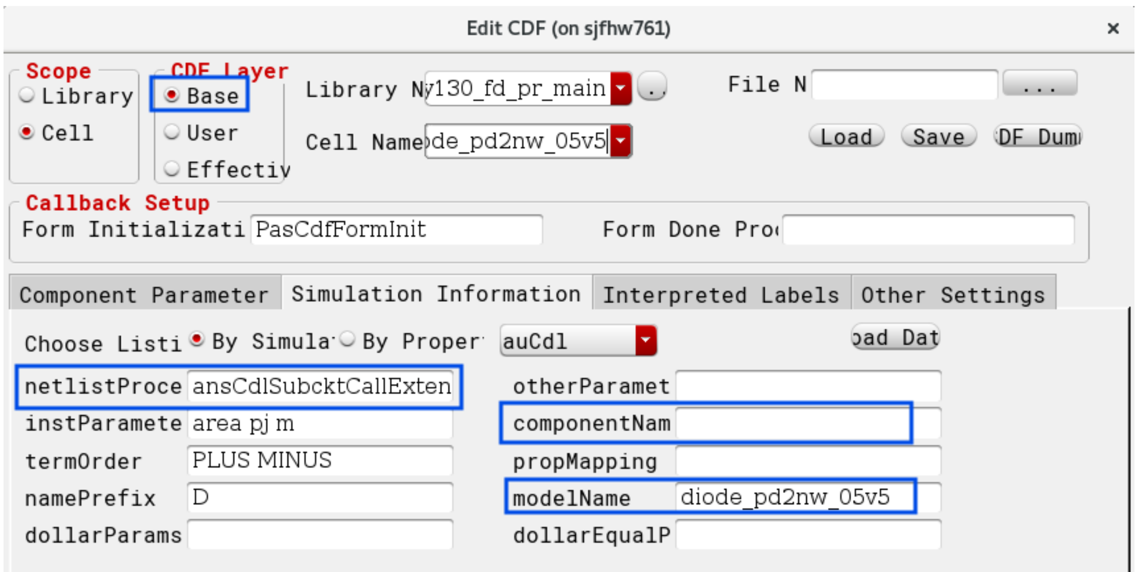
Change required: CDF

Under Simulation Information, auCDL must be updated:

netlistProcedure: ansCdlSubcktCallExtended

modelName: *SHOULD MATCH cellname (see example)*

componentName: should be empty



Cell Type: vpp capacitors

Changes are required to all CDFs and some layouts; details below.
All VPP cap edits (CDF and layouts) can be found in the "cap_vpp_fixes.tar" file; un-tar this file inside the directory of the PDK library: sky130_release_0.0.#/libs/sky130_fd_pr_main

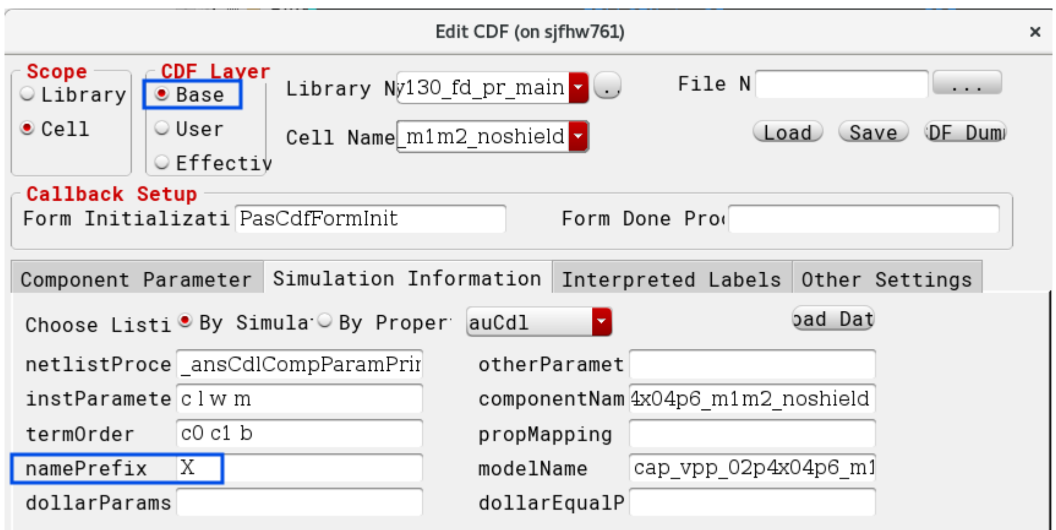
CDF Changes

List of cells affected: All "cap_vpp_XXX" cells

    Example: cap_vpp_02p4x04p6_m1m2_noshield

Change required:

    Under Simulation Information, auCDL "namePrefix" must be "X"



Layout Changes

"top" pin must be added to the layouts of some of the cap_vpp cells:

cap_vpp_04p4x04p6_l1m1m2_shieldpo_floatm3
cap_vpp_04p4x04p6_m1m2m3_shieldl1
cap_vpp_04p4x04p6_m1m2m3_shieldl1m5_floatm4
cap_vpp_06p8x06p1_l1m1m2m3_shieldpom4
cap_vpp_06p8x06p1_m1m2m3_shieldl1m4
cap_vpp_08p6x07p8_l1m1m2_shieldpo_floatm3
cap_vpp_08p6x07p8_m1m2m3_shieldl1m5_floatm4
cap_vpp_11p5x11p7_l1m1m2m3m4_shieldm5
cap_vpp_11p5x11p7_l1m1m2m3m4_shieldpom5
cap_vpp_11p5x11p7_l1m1m2m3_shieldm4
cap_vpp_11p5x11p7_l1m1m2m3_shieldpom4
cap_vpp_11p5x11p7_l1m1m2_shieldpom3
cap_vpp_11p5x11p7_m1m2m3m4_shieldl1m5
cap_vpp_11p5x11p7_m1m2m3m4_shieldm5
cap_vpp_11p5x11p7_m1m2m3_shieldl1m5_floatm4

Here is an example for cell “cap_vpp_04p4x04p6_m1m2m3_shieldl1”